



VPC

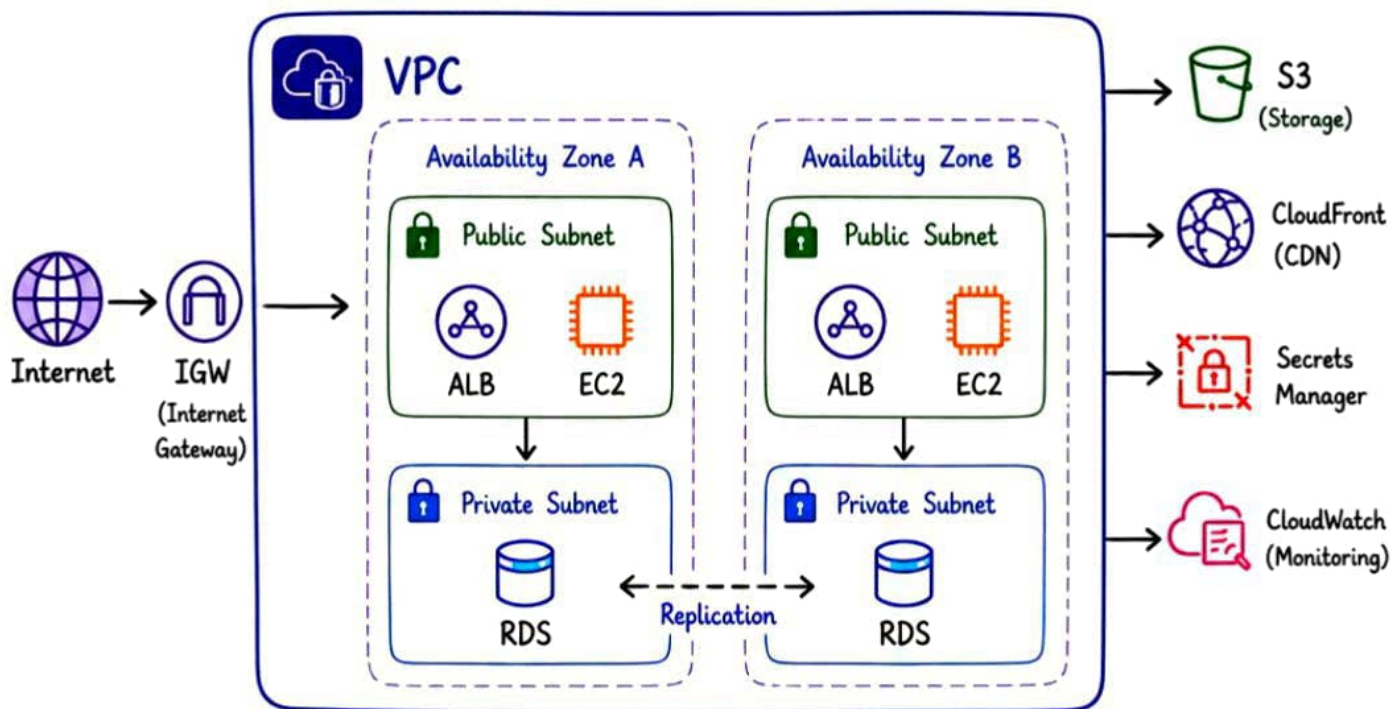
VIRTUAL PRIVATE CLOUD

BUILD • ISOLATE • SECURE • SCALE



VPC is the foundation of your AWS cloud.

Isolate your resources. **Secure** your network. **Scale** your architecture.



LEGEND

Public Subnet (Internet facing)	ALB (Application Load Balancer)
Private Subnet (Internal)	EC2 (Compute Instance)
	RDS (Relational Database Service)
	IGW (Internet Gateway)

WHAT YOU WILL LEARN

☒ VPC Basics & Components	☒ VPC Endpoints
☒ Subnets & CIDR	☒ Peering & Transit Gateway
☒ Routing & Route Tables	☒ On-Premises Connectivity
☒ Security (SG & NACL)	☒ Architecture & Troubleshooting



A STRONG NETWORK TODAY,
A **SECURE & SCALABLE** ARCHITECTURE TOMORROW!



AWS VPC

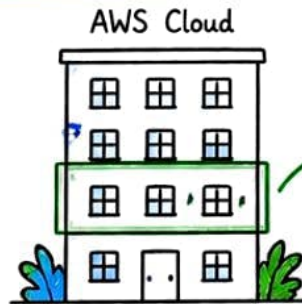
②

1. WHAT IS VPC?

Amazon VPC (Virtual Private Cloud) is a logically isolated virtual network in the AWS Cloud where you can launch and manage your resources.

Analogy:

Think of VPC as your own private space in a big apartment building. You get your own rooms, walls, doors and rules.



Your VPC

- ✓ Your space
- ✓ Your rules
- ✓ Your resources
- ✓ Your network

2. WHY DO WE NEED VPC?

- **Isolation** → Keeps your resources private and separated from others.
- **Control** → You control IP ranges, subnets, routing, gateways and security.
- **Security** → Apply your own security (Security Groups, NACLs, etc.).
- **Custom Networking** → Design the network the way your application needs.
- **Scalability** → Easily scale your network and resources as you grow.
- **Cost Optimization** → Use only what you need and design efficiently.

3. KEY CONCEPTS IN VPC

REGION



A Region is a geographical area where AWS has multiple data centers.

AVAILABILITY ZONE (AZ)



Each Region has multiple isolated locations called Availability Zones.

CIDR BLOCK

10.0.0.0/16

A range of IP addresses that you assign to your VPC.

VPC



Your Virtual Private Cloud – your own isolated network in the AWS Cloud.



IN SHORT

VPC gives you a **private**, **secure** and fully **controlled** network environment in the AWS Cloud.



VPC CIDR & IP ADDRESSING

3

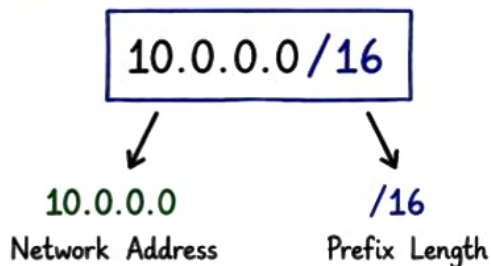
WHAT IS CIDR?

CIDR (Classless Inter-Domain Routing) is a method used to allocate IP addresses more efficiently.



In VPC, CIDR defines the IP address range for your VPC.

CIDR NOTATION



- The /16 means first 16 bits are for the network.
- Remaining 16 bits are for hosts.
- Total IPs = $2^{16} = 65,536$
- Usable IPs = $65,536 - 5 = 65,531$ (AWS reserves 5 IPs)

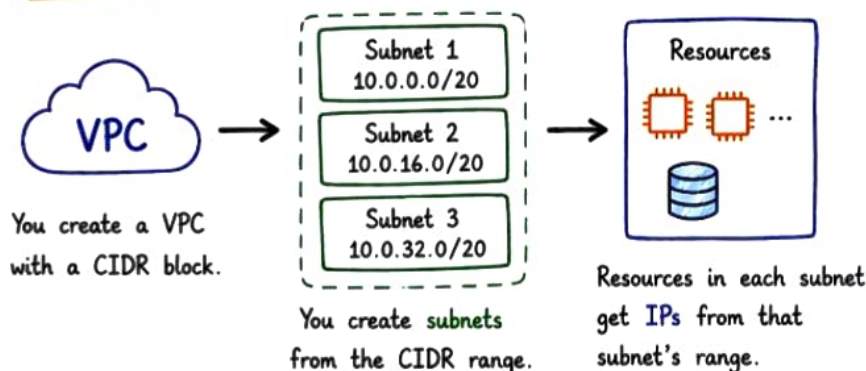
COMMON CIDR BLOCKS IN VPC

CIDR Block	Total IPs	Usable IPs (Approx.)	Use Case
10.0.0.0/16	65,536	65,531	Large networks
10.0.0.0/20	4,096	4,091	Medium networks
10.0.0.0/24	256	251	Small networks
10.0.0.0/28	16	11	Very small networks

AWS RESERVES 5 IPs

- Network address
- VPC router
- DNS
- Future use
- Broadcast address

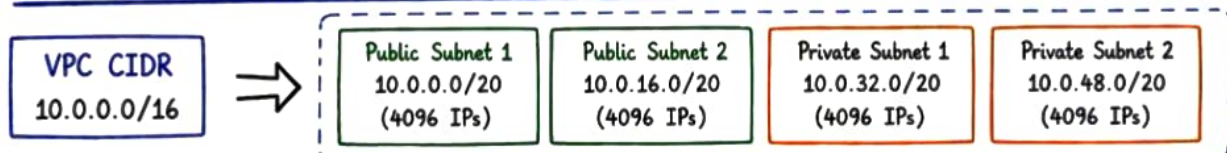
HOW IPs ARE ASSIGNED IN VPC



POINTS TO REMEMBER

- ✓ Choose a CIDR block that **does not overlap** with your on-premise network.
- ✓ Plan your IP ranges carefully for **future growth**.
- ✓ You **cannot change** the CIDR block of a VPC after creation.

EXAMPLE - VPC AND SUBNET IP PLANNING



★ This is just an example. You can design as per your architecture and needs.



VPC SUBNETS

4

WHAT IS A SUBNET?

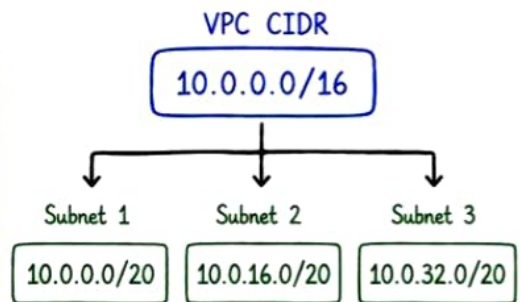
A Subnet is a range of IP addresses within your VPC CIDR block.
It lets you divide your VPC into smaller networks and place your resources in them.



Subnets help you organize your resources based on role, security and access needs.

KEY POINTS

- ★ Subnets exist inside a VPC.
- ★ Each subnet must be in one Availability Zone (AZ).
- ★ You can create multiple subnets across multiple AZs.
- ★ Each subnet has its own CIDR block (smaller range).
- ★ You cannot overlap CIDR blocks of subnets in the same VPC.



Subnets are carved out from the VPC CIDR range.

SUBNET TYPES (BY DEFAULT BEHAVIOR)

1. PUBLIC SUBNET

- Has a route to the Internet Gateway (IGW)
- Resources can be accessed from the Internet (if allowed by SG).
- Example: ALB, Bastion Host, NAT Gateway



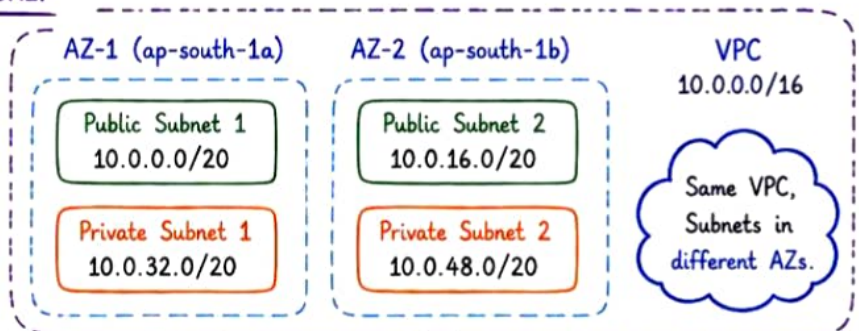
2. PRIVATE SUBNET

- No direct route to the Internet.
- Resources cannot be accessed from the Internet directly.
- Example: App servers, Databases



SUBNET AND AZ RELATIONSHIP

- Each subnet belongs to one Availability Zone.
- To build a highly available architecture, create subnets in multiple AZs.



Subnets are the foundation for placing and isolating your resources in the right place with the right access.





PUBLIC vs PRIVATE SUBNET

5

WHAT MAKES A SUBNET PUBLIC OR PRIVATE?

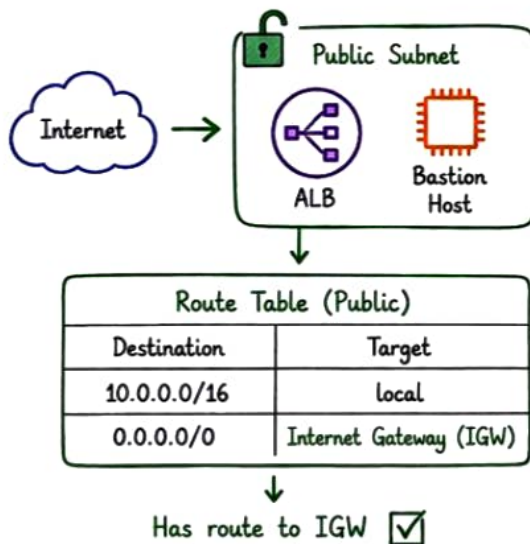
A subnet is **not** public or private by itself. It depends on its **route table**.



It's the route to the Internet that decides whether a subnet is public or private.

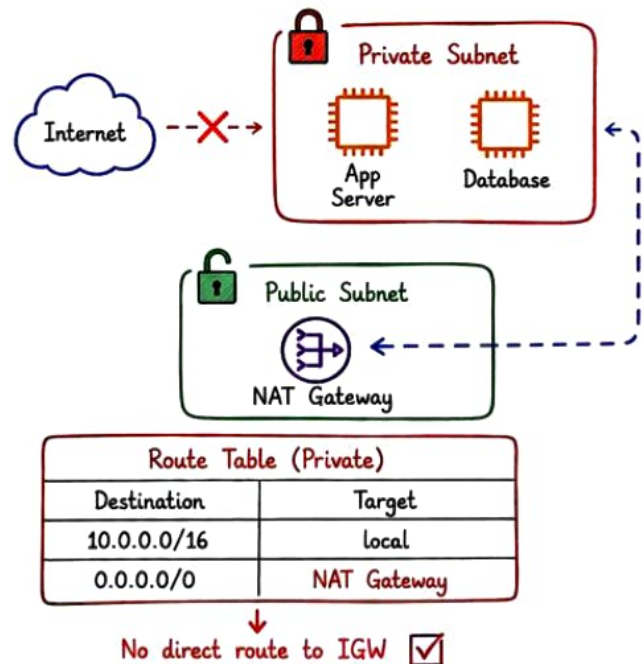
PUBLIC SUBNET

- Has a route to the Internet Gateway (IGW).
- Resources in this subnet can be accessed from the Internet (if allowed by SG).
- Used for resources that need to be publicly accessible.
- Examples: ALB, Bastion Host, NAT Gateway (if placed in public subnet).



PRIVATE SUBNET

- No direct route to the Internet.
- Resources cannot be accessed from the Internet directly.
- Used for backend resources like App servers, Databases, etc.
- Internet access (if needed) is through NAT Gateway in public subnet.



QUICK COMPARISON

Feature	Public Subnet	Private Subnet
Internet route	Has route to IGW	No route to IGW (uses NAT)
Internet access	Accessible from Internet (if allowed by SG)	Not accessible from Internet directly
Used for	Public facing resources	Backend / internal resources
Examples	ALB, Bastion Host, NAT Gateway	App Servers, Databases

★ IMPORTANT

- Public ≠ Less Secure
Security is controlled by Security Groups and NACLs.
- Private ≠ No Internet
Private subnets can access the Internet using NAT Gateway.



Rule of Thumb:

Put what the world needs to access in Public Subnets.

Keep what the world doesn't need to see in **Private Subnets**.





VPC COMPONENTS

6

These are the building blocks that make a VPC work.



Think of components as tools, and VPC as your toolbox.

1. INTERNET GATEWAY (IGW)



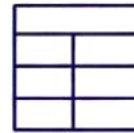
Enables communication between resources in your VPC and the Internet.

2. NAT GATEWAY



Allows resources in private subnets to access the Internet for outbound traffic (e.g., updates).

3. ROUTE TABLE



A set of rules that determines where network traffic is directed.

4. SECURITY GROUP (SG)



Acts as a virtual firewall for your instances. It is stateful and instance-level.

5. NETWORK ACL (NACL)



Acts as a firewall for your subnets. It is stateless and works at the subnet level.

6. VPC ENDPOINT



Allows private connection to AWS services without using the Internet.

7. DHCP OPTIONS SET



Provides IP address range and DNS server information to resources in your VPC.

8. VPC PEERING / TRANSIT GATEWAY



VPC Peering connects two VPCs.

Transit Gateway connects many VPCs and on-premises networks.

SECURITY GROUP vs NACL

Feature	Security Group	Network ACL
Scope	Instance level	Subnet level
Stateful / Stateless	Stateful	Stateless
Rule Allow / Deny	Allow only	Allow and Deny
Default	All Deny	All Allow
Rule Evaluation	First match	Numbered order
Return Traffic	Automatically allowed	Must allow explicitly

HOW TRAFFIC FLOW WORKS?



- 1 Traffic comes from/to the Internet.
- 2 IGW allows the traffic into the VPC.
- 3 NACL (inbound) rules are checked.
- 4 Security Group rules are checked.
- 5 If allowed, traffic reaches the instance.
- 6 For response traffic, return path is automatically allowed in SG (but must be allowed in NACL).

VPC ENDPOINT TYPES

Gateway Endpoint		Used for S3, DynamoDB. Route table based.
Interface Endpoint		Used for most AWS services. Works with Private IP (ENI in your subnet).



KEY TAKEAWAY

- Use **Public Subnets** for resources that need Internet access.
- Use **Private Subnets** for resources that should not be directly accessible from the Internet.
- Use **NAT Gateway** for private subnets to access the Internet.
- Use **Security Groups** and **NACLs** for security.



Understand these components well - they are the foundation of every VPC design!



VPC ROUTING

7



WHAT IS ROUTING?

Routing is the process of deciding the path for network traffic to reach its destination.



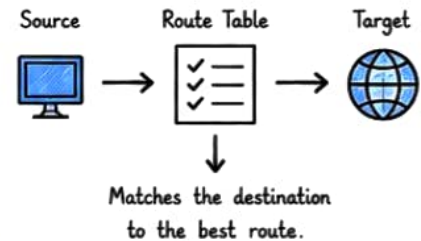
Think of Route Table as a GPS that tells your traffic where to go.



DEFAULT ROUTES IN A VPC

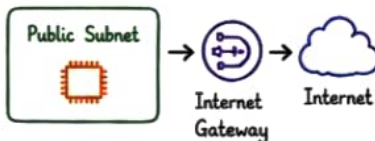
Destination	Target	Purpose
10.0.0.0/16	local	Traffic within the VPC stays inside (local).
0.0.0.0/0	Internet Gateway (IGW)	Traffic to the Internet.
0.0.0.0/0	NAT Gateway	Traffic from private subnet to Internet.

HOW ROUTING WORKS



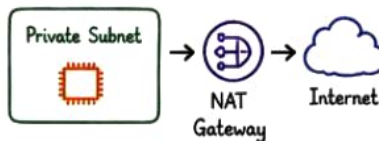
EXAMPLE ROUTING SCENARIOS

① Internet access from Public Subnet



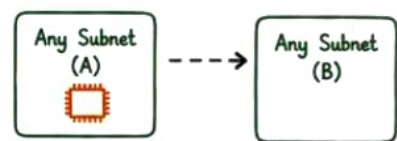
- Route Table has 0.0.0.0/0 → IGW
- Instance can access the Internet.
- Internet can access the instance (if SG allows).

② Internet access from Private Subnet



- Route Table has 0.0.0.0/0 → NAT GW
- Instance can access the Internet (outbound only).
- Internet cannot initiate connection to the instance.

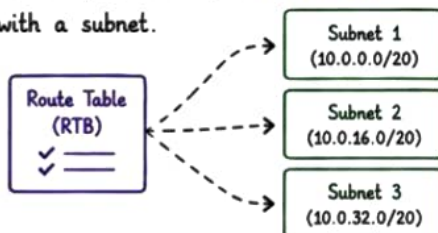
③ Internal communication within VPC



- Default route 10.0.0.0/16 → local
- Traffic stays inside the VPC.
- No Internet Gateway or NAT needed.

ROUTE TABLE ASSOCIATION

A route table must be associated with a subnet.



★ One subnet can be associated with only one route table at a time.

ROUTE TABLE TYPES

Main Route Table

- Created automatically with the VPC.
- Associated with subnets by default (if no other association).
- You cannot delete the main route table.



Custom Route Table

- Created by you.
- You can associate with specific subnets.
- You can have multiple custom route tables.



BEST PRACTICES

- ✓ Use least privilege in Security Groups and NACLs.
- ✓ Use NAT Gateway for private subnet Internet access.
- ✓ Keep route tables simple and easy to understand.



- ✓ Use meaningful naming for route tables.
- ✓ Always verify route table association after changes.
- ✓ Test connectivity after any routing change.



Good routing design = Secure, Reliable and Efficient Network!



VPC SECURITY

8



Security in a VPC is managed using Security Groups and NACLs.



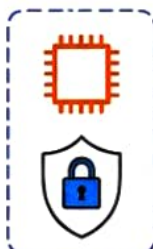
Think of SG as **Lock** on the door and NACL as Security at the gate.



1. SECURITY GROUP (SG)

- Acts at instance level.
- **Stateful**: If you allow inbound, response is automatically allowed.
- Allow rules only (no deny rule).
- Evaluated at the instance ENI level.
- **Default**: All inbound denied, all outbound allowed.

Instance



2. NETWORK ACL (NACL)

- Acts at subnet level.
- **Stateless**: You must allow both inbound and outbound.
- Allow and **Deny** rules.
- Evaluated before traffic reaches the instance.
- **Default**: All inbound and outbound denied.

Subnet



SECURITY GROUP vs NACL

Feature	Security Group (SG)	Network ACL (NACL)
Level	Instance level (ENI)	Subnet level
Stateful / Stateless	Stateful	Stateless
Rule Type	Allow only	Allow and Deny
Rule Direction	Inbound rules only (outbound auto-allowed)	Inbound and Outbound rules
Default	All inbound denied, outbound allowed	All inbound denied, outbound denied
Evaluation Order	After NACL	Before Security Group
Use Case	Control traffic to instances	Control traffic at subnet boundary
Max Rules	60 rules (per SG)	20 rules (inbound) + 20 rules (outbound)

EXAMPLE RULES

SECURITY GROUP (Instance Level)

Type	Protocol	Port	Source	Description
Inbound	TCP	80	0.0.0.0/0	Allow HTTP
Inbound	TCP	22	My IP /32	Allow SSH
Outbound	All	All	0.0.0.0/0	All Outbound

- ★ No explicit deny rule needed. Return traffic is automatically allowed.

NETWORK ACL (Subnet Level)

Rule #	Type	Protocol	Port Range	Source / Dest	Allow / Deny
100	Inbound	TCP	80	0.0.0.0/0	ALLOW
110	Inbound	TCP	22	My IP /32	ALLOW
120	Inbound	All	All	0.0.0.0/0	DENY
200	Outbound	All	All	0.0.0.0/0	ALLOW
*	Outbound	All	All	0.0.0.0/0	DENY

- ★ Rule numbers evaluated in order (lowest first). Add explicit rules for both directions.

★ KEY TAKEAWAYS

- ✓ Use NACLs for subnet-level security (coarse control).
- ✓ Use Security Groups for instance-level security (fine control).
- ✓ NACLs are stateless - be explicit.
- ✓ Security Groups are stateful - easier to manage.
- ✓ Layer both for best security.

BEST PRACTICES

- ① Keep NACLs simple. Use them to block unwanted ranges.
- ② Use Security Groups to allow only required ports.
- ③ Always restrict SSH (22) to your IP.
- ④ Use least privilege principle.
- ⑤ Regularly review and audit your rules.
- ⑥ Test connectivity after any rule change.



Good security is not about blocking everything, it's about allowing the right thing!



VPC FLOW (TRAFFIC PATH)

9



HOW DOES TRAFFIC FLOW IN A VPC?

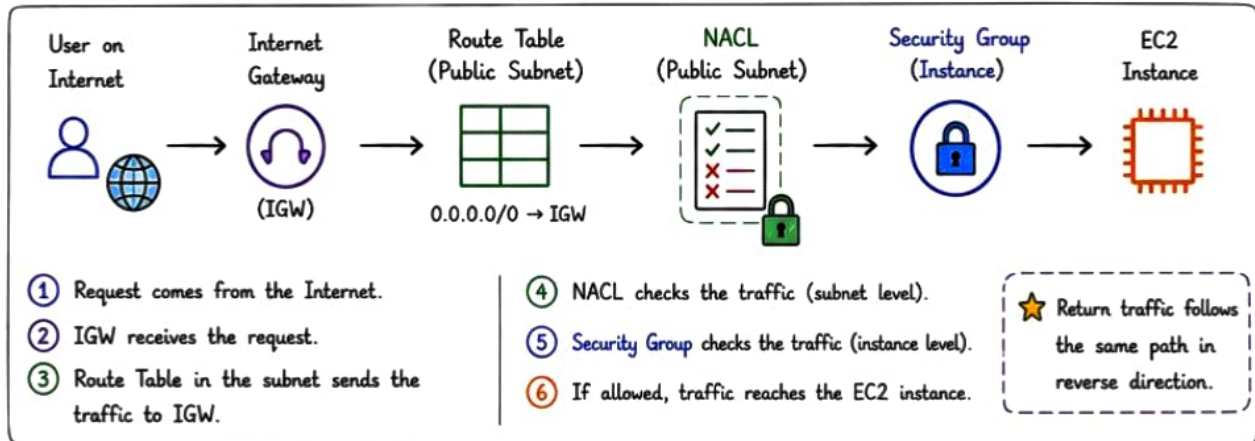
Traffic follows a specific path based on source, destination and the rules in Route Table, NACL and Security Group.



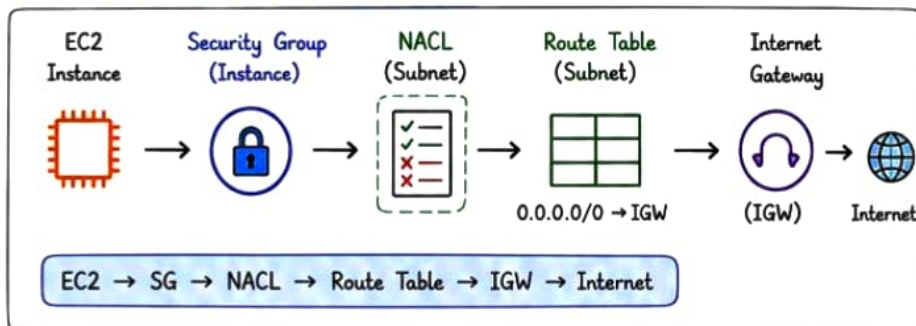
Always remember the order:
NACL (Subnet) → SG (Instance)
→ Reach Destination



END-TO-END TRAFFIC FLOW (INTERNET TO EC2 INSTANCE)



OUTBOUND TRAFFIC (EC2 TO INTERNET)



KEY POINTS

- ★ NACL is stateless (check both inbound and outbound).
- ★ Security Group is stateful (return traffic is auto-allowed).
- ★ Route Table decides where traffic goes.

INTERNET GATEWAY (IGW) vs NAT GATEWAY

	Internet Gateway (IGW)	NAT Gateway
Used By	Public Subnets	Private Subnets
Purpose	Internet access for resources in public subnets	Internet access for resources in private subnets
Allows	Inbound + Outbound	Outbound only
Direction	Bidirectional	Private → Internet
Public IP	Not required	Required (Elastic IP)
Location	Internet Gateway (attached to VPC)	In a Public Subnet

COMMON PORTS TO REMEMBER

Service / Use	Protocol	Port	Direction	Where Used
HTTP	TCP	80	In / Out	Web Traffic
HTTPS	TCP	443	In / Out	Secure Web Traffic
SSH	TCP	22	In	Server Access
Custom App	TCP	3000	In / Out	Application
MySQL	TCP	3306	In	Database
RDP	TCP	3389	In	Windows Access

★ Always open **only** the required ports. Close everything else!



TROUBLESHOOTING CHECKLIST

- 1 Check Route Table
 Is the destination route correct?
- 2 Check NACL
 Are inbound and outbound rules allowing traffic?
- 3 Check Security Group
 Is the required port allowed? Source IP correct?
- 4 Check Destination
 Is the service running on the correct port?
- 5 Test Connectivity
 Use ping, telnet, nc, curl, etc.



Understand the flow, follow the order, and your VPC will always work like magic!





VPC SUBNETS

10



WHAT IS A SUBNET?

A subnet is a range of IP addresses in your VPC. It allows you to organize resources and control how traffic flows.



Think of Subnets as rooms inside your VPC building.



SUBNET TYPES



PUBLIC SUBNET

- Has a route to Internet Gateway (IGW).
- Resources can be accessed from the Internet.



PRIVATE SUBNET

- No direct route to Internet.
- Used for backend resources like App servers, Databases.

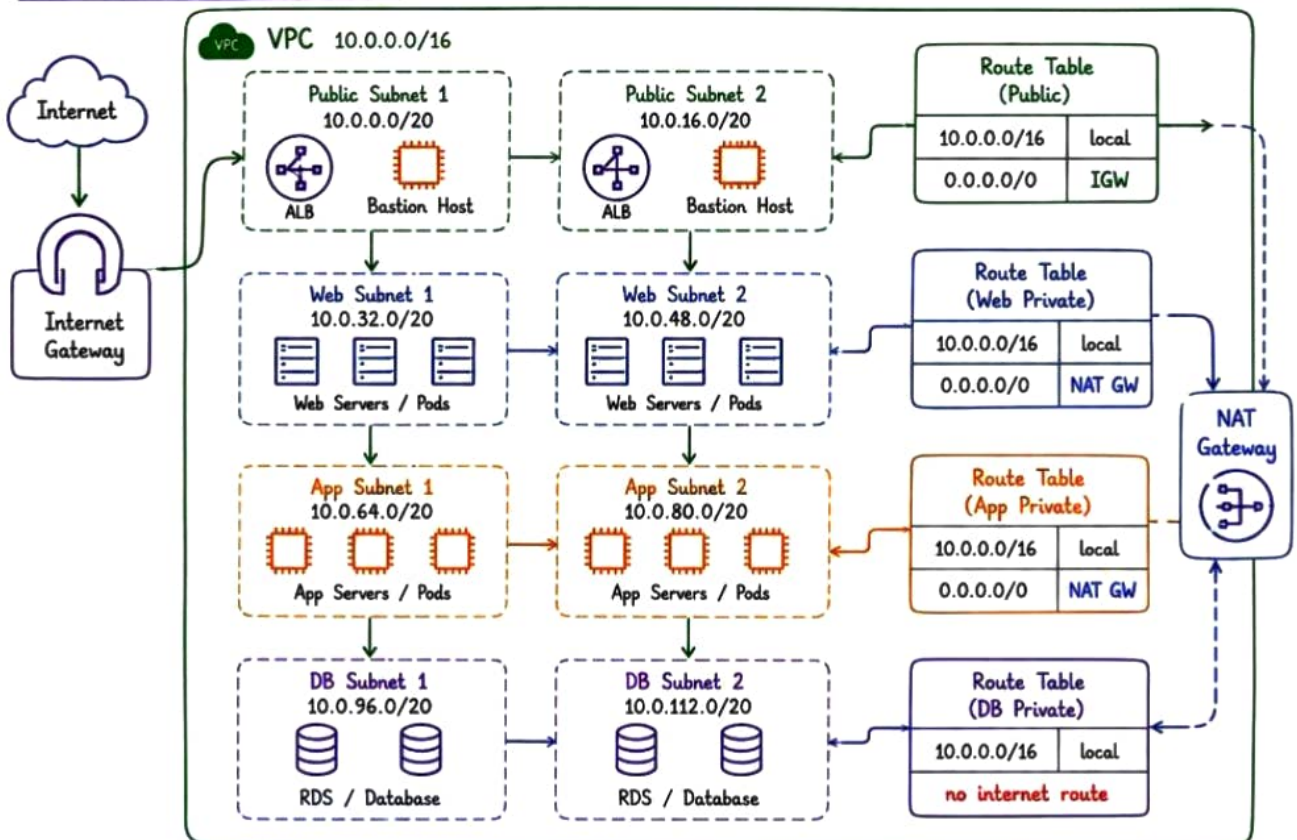


OUR VPC SUBNET DESIGN

Type	Name	CIDR Block	AZ (Example)	Purpose
Public	Public Subnet 1	10.0.0.0/20	ap-south-1a	ALB, Bastion, IGW
Public	Public Subnet 2	10.0.16.0/20	ap-south-1b	ALB, Bastion, IGW
Private (Web)	Web Subnet 1	10.0.32.0/20	ap-south-1a	Web Servers / Pods
Private (Web)	Web Subnet 2	10.0.48.0/20	ap-south-1b	Web Servers / Pods
Private (App)	App Subnet 1	10.0.64.0/20	ap-south-1a	App Servers / Pods
Private (App)	App Subnet 2	10.0.80.0/20	ap-south-1b	App Servers / Pods
Private (DB)	DB Subnet 1	10.0.96.0/20	ap-south-1a	RDS / Database
Private (DB)	DB Subnet 2	10.0.112.0/20	ap-south-1b	RDS / Database

★ We are using /20 subnets (4096 IPs each).

HOW SUBNETS WORK TOGETHER



★ KEY POINTS

- Public subnets have a route to IGW.
- Private subnets use NAT Gateway for outbound internet.
- DB subnets have NO internet route for security.
- Each subnet has its own route table.

BEST PRACTICES

- ✓ Keep public subnets only for ALB, Bastion, NAT.
- ✓ Place Web, App and DB in separate private subnets.
- ✓ Use at least 2 subnets per tier across AZs for High Availability.
- ✓ Keep route tables simple and purpose-based.



Good subnet design = Better Security, High Availability and Scalable Architecture!



VPC ENDPOINTS

11



WHAT IS A VPC ENDPOINT?

A VPC Endpoint enables private connectivity between your VPC and AWS services without using the Internet, NAT Gateway or IGW.



Think of Endpoints as a private door to AWS services inside your VPC.



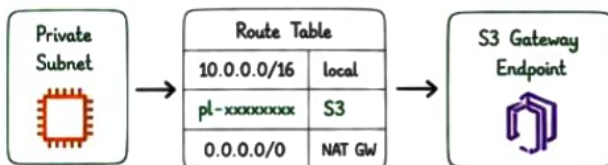
WHY USE VPC ENDPOINTS?

 More Secure Traffic stays within AWS network.	 Cost Effective Avoid data transfer charges via IGW or NAT.	 Better Performance Lower latency, more reliable connections.	 Access Control Control access using Security Groups and policies.
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TYPES OF VPC ENDPOINTS

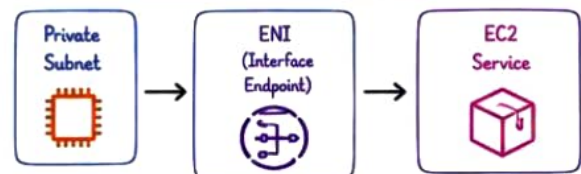
Type	Connects To	Access	Uses	Example Services	Network Interface
Gateway Endpoint 	S3, DynamoDB	Route Table (Subnet Level)	Public and Private Subnets	Amazon S3 DynamoDB	No
Interface Endpoint 	Most AWS Services	Security Group (ENI Level)	Private Subnets Only	EC2, CloudWatch, Secrets Manager, SSM, ECR, etc.	Yes (ENI in your subnet)
Gateway Load Balancer Endpoint 	Your GWLBs	Route Table (Subnet Level)	Private Subnets Only	3rd Party Virtual Appliances	No

HOW GATEWAY ENDPOINT WORKS (S3 EXAMPLE)



- Route Table has a prefix list for S3.
- Traffic goes to S3 over AWS backbone (no Internet).

HOW INTERFACE ENDPOINT WORKS (EC2 EXAMPLE)



- An ENI is created in your subnet.
- Your resources connect to the service using a private IP.

COMMON USE CASES

 S3 Gateway Endpoint	➔ Private access to S3 for backups, logs, static website content.
 DynamoDB Gateway Endpoint	➔ Private access to DynamoDB tables.
 CloudWatch Interface Endpoint	➔ Send metrics and logs privately from private subnets.
 Secrets Manager Interface Endpoint	➔ Retrieve secrets securely without using Internet or NAT.
 SSM Interface Endpoint	➔ Use Session Manager, Patch Manager from private subnets.

ENDPOINT COMPARISON

Feature	Gateway Endpoint	Interface Endpoint
Supported Services	S3, DynamoDB	Most AWS Services
Access Method	Route Table	ENI + Security Group
Subnet Support	Public + Private	Private Only
Network Interface	No	Yes
IP Address	Not applicable	Private IP in Subnet
Cost	Free	Hourly + Data Processing
Use Case	High throughput (AWS storage)	Fine-grained access (Other services)

BEST PRACTICES

- ✓ Use Gateway Endpoints for S3 and DynamoDB.
- ✓ Use Interface Endpoints for other services used in private subnets.
- ✓ Restrict access using Security Groups and IAM policies.
- ✓ Keep your route tables clean and purpose-based.
- ✓ Monitor endpoint usage with VPC Flow Logs and CloudWatch.



★ KEY TAKEAWAY

- VPC Endpoints keep traffic private, secure and cost-effective.
- Gateway Endpoints = Route Table based (S3, DynamoDB).
- Interface Endpoints = ENI based (Most Services).
- Use the right endpoint for the right service!



Private connectivity is powerful - VPC Endpoints make your cloud network stronger!



VPC PEERING & TRANSIT GATEWAY

12



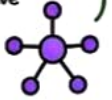
WHAT IS VPC PEERING?

VPC Peering lets you connect two VPCs privately using AWS backbone network.
Traffic stays private (no Internet, no NAT).

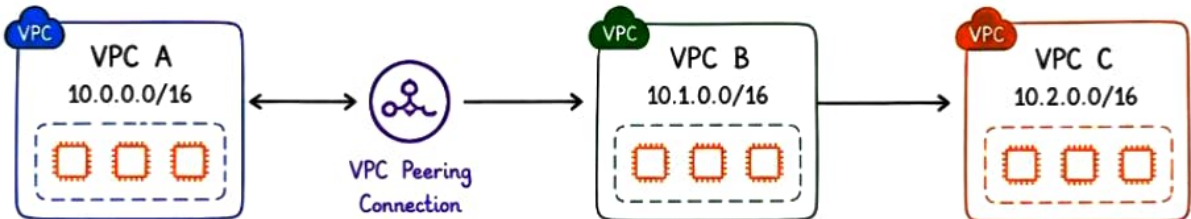


Use Peering for simple connections between few VPCs.

Use Transit Gateway when you have many VPCs or hybrid network.



1. VPC PEERING (POINT-TO-POINT CONNECTION)



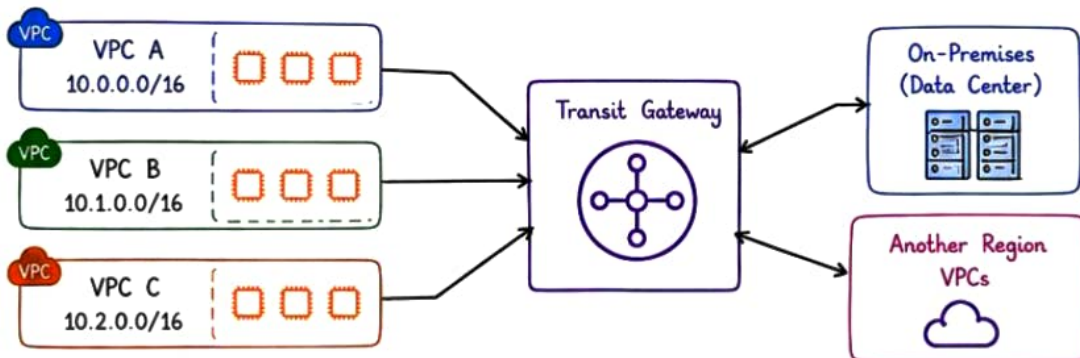
- Each connection is between two VPCs.
- You need a peering connection for every pair.
- Route tables in both VPCs must be updated.
- Does not support **transitive routing**.



Best For :

Connecting 2-3 VPCs in same or different regions.

2. TRANSIT GATEWAY (HUB-AND-SPOKE)



- All VPCs connect to a central Transit Gateway.
- One connection to TGW, not to every VPC.
- Supports **transitive routing**.
- Easy to scale and manage.



Best For :

Large and complex architectures, multiple VPCs, hybrid networks.

PEERING vs TRANSIT GATEWAY

VPC PEERING

- Point-to-point connection
- Not scalable
- No transitive routing
- Lower cost
- Simple use cases

TRANSIT GATEWAY

- Hub-and-spoke model
- Highly scalable
- Supports transitive routing
- Slightly higher cost
- Complex / large use cases

BEST PRACTICES

- ✓ Keep route tables simple and specific.
- ✓ Use meaningful CIDR blocks to avoid overlap.
- ✓ Use Transit Gateway for scalability.
- ✓ Monitor connections using VPC Flow Logs.
- ✓ Test connectivity after creating routes.



COMMON USE CASES



Multiple VPCs in same Region → Use Transit Gateway



VPCs in different Regions → Use Transit Gateway



Connect to On-Premises → Use Transit Gateway



Two VPCs need to talk → Use VPC Peering



Peering is for **few**, Transit Gateway is for **many**!



VPC DNS & ROUTE 53 RESOLVER

13



WHY DNS IN VPC?

- DNS (Domain Name System) translates domain names like `app.example.com` into IP addresses.
- Route 53 Resolver is AWS managed DNS service that handles DNS queries inside your VPC.



Think of DNS as phonebook and Resolver as the service that looks up the number for you!



HOW DNS WORKS IN VPC (BASIC FLOW)



- Your application uses the IP to connect to the service.

AMAZON PROVIDED DNS IN VPC

- Every VPC has a default DNS resolver at the magic IP: **169.254.169.253**
- It is enabled by default when you create a VPC.
- Available in all subnets in the VPC.

VPC DNS SETTINGS (Per VPC)

- ☒ **Enable DNS hostnames:** Allows Amazon to assign DNS names to instances.
- ☒ **Enable DNS resolution:** Allows DNS queries to be resolved using Route 53 Resolver.

COMPONENTS OF ROUTE 53 RESOLVER



- 1. Resolver (Amazon Provided)**
Default resolver (169.254.169.253) that forwards queries on your behalf.
- 2. Resolver Endpoint**
Enables DNS queries from on-premises networks or other VPCs to forward to Route 53 Resolver.
- 3. Resolver Rule**
Defines which DNS queries are forwarded to outbound endpoints (forwarding rules) or inbound endpoints (inbound rules).

PRIVATE DNS vs PUBLIC DNS

- Public DNS**
- Used for Internet-facing resources.
- Resolvable over the Internet.

- Private DNS**
- Used for internal resources within VPC.
- Not resolvable over the Internet.

PRIVATE DNS IN ROUTE 53

- When you create a record in Route 53 Private Hosted Zone, it is only resolvable within the VPC.
- Use Private Hosted Zones for internal domain names like `app.internal` or `db.company.local`.
- It does not require Internet access.



COMMON USE CASES

- Resolve AWS resources** → Resolve EC2, RDS, ELB, S3 (via VPC Endpoint) names.
- On-premises to AWS DNS** → Use Resolver Endpoints to forward DNS queries from on-premises to VPC.
- Cross VPC DNS Resolution** → Share DNS across VPCs using Resolver Endpoints.
- Hybrid Cloud** → Use Route 53 Resolver for consistent name resolution across on-prem and AWS.

BEST PRACTICES

- ☒ Keep DNS hostnames and DNS resolution enabled.
- ☒ Use Private Hosted Zones for internal apps.
- ☒ Use Resolver Endpoints for hybrid connectivity.
- ☒ Use meaningful domain names.
- ☒ Monitor DNS queries using Route 53 Resolver Query Logging (to CloudWatch Logs).



DNS makes names easy, Resolver makes it possible inside your VPC!



VPC TO ON-PREMISES CONNECTIVITY

14



WHY CONNECT ON-PREMISES?

Many organizations need to connect their VPC with on-premises data center for secure access to applications, databases, files, printers, etc.

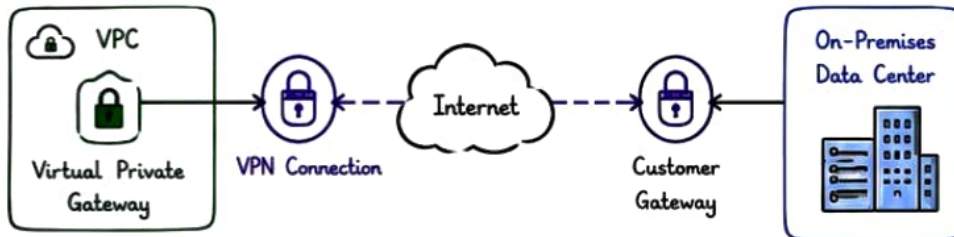


Two main ways to do this in AWS:

1. VPN Connection
2. AWS Direct Connect



1. VPN CONNECTION (OVER THE INTERNET)



Pros:

- ✓ Easy to set up
- ✓ Cost effective
- ✓ Uses Internet

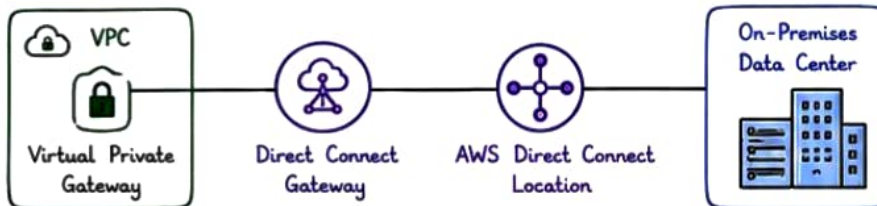
Cons:

- ✓ Depends on Internet
- ✓ Lower bandwidth
- ✓ Not ideal for high throughput

How it works:

- You create a Virtual Private Gateway (VGW) in your VPC.
- Create a Customer Gateway (CGW) on your on-premises side.
- Establish a VPN tunnel between VGW and CGW over the Internet.

2. AWS DIRECT CONNECT (PRIVATE CONNECTION)



Pros:

- ✓ High bandwidth
- ✓ More secure (private network)
- ✓ Consistent performance

Cons:

- ✓ Higher cost
- ✓ Takes more time to set up
- ✓ Requires physical connection

How it works:

- You establish a physical connection from your data center to an AWS Direct Connect location.
- From there, private connection to your VPC via Direct Connect Gateway.

3. VPN vs DIRECT CONNECT

Feature	VPN Connection	AWS Direct Connect
Connection Type	Over Internet	Private Network
Setup Time	Minutes to hours	Days to weeks
Cost	Low	High
Bandwidth	Up to ~1.25 Gbps (per tunnel)	Up to 100 Gbps or more
Reliability	Depends on Internet	Very Reliable
Use Case	Small offices, Dev/Test	Production, High data transfer

Important Note



For more availability and resilience, use:

- VPN: Create 2 tunnels (different ISPs if possible)
- Direct Connect: Use Redundant Connections (primary + secondary)

USE CASES



Hybrid Workload

Access on-prem databases from applications running in AWS.



Data Backup & DR

Backup data from on-prem to AWS or replicate for Disaster Recovery.



Centralized Management

Manage users, syslog, monitoring from your on-prem tools.



Choose the right connection based on your **needs**, **budget** and **performance**!



COMPLETE VPC ARCHITECTURE

15

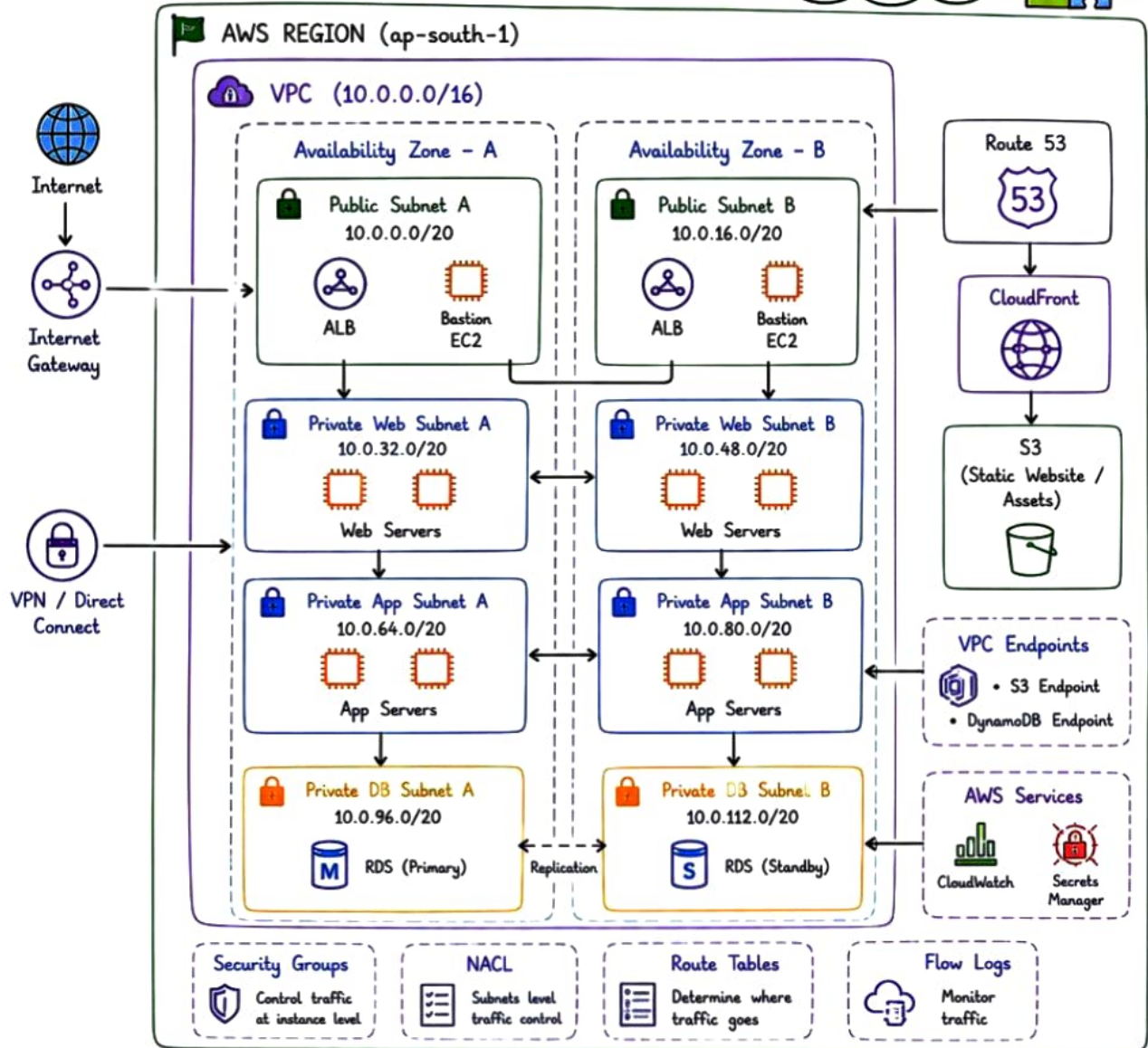


PUTTING IT ALL TOGETHER!

A well-designed VPC architecture is secure, scalable, highly available and cost-effective.



This is how all the pieces we learned fit together!



KEY HIGHLIGHTS



- ✓ Multi-AZ for High Availability
- ✓ Public subnets for Internet facing resources only
- ✓ Private subnets for application & database tiers
- ✓ Least privilege access using Security Groups and NACLs
- ✓ Private connectivity using Endpoints, VPN or Direct Connect
- ✓ Monitoring with Flow Logs and CloudWatch

TRAFFIC FLOW (Example)

- 1 User → Route 53 → CloudFront / ALB
- 2 ALB → Web Servers (Private Subnets)
- 3 Web Servers → App Servers (Private Subnets)
- 4 App Servers → RDS (Private DB Subnets)
- 5 App Servers → S3 / Other AWS Services via VPC Endpoints (No Internet)

Traffic stays within AWS network as much as possible!



BENEFITS

- ✓ Secure - No direct access to private resources.
- ✓ Scalable - Add more subnets, instances easily.
- ✓ Reliable - Multi-AZ design.
- ✓ Performant - Local routing within AWS network.
- ✓ Cost Effective - Use Private Endpoints, NAT Gateway only when needed.



A well-architected VPC is the foundation of a secure and scalable cloud!



VPC TROUBLESHOOTING

16



WHY TROUBLESHOOTING MATTERS?

Even in a well-designed VPC, issues can happen. Knowing how to troubleshoot quickly saves time and keeps your applications running!



General Approach:

1. Identify the problem
2. Check the path (Traffic Flow)
3. Verify configuration & permissions
4. Test and fix



COMMON VPC ISSUES & HOW TO FIX

Issue	Symptoms	Possible Causes	How to Fix
 Instance cannot reach Internet	- No internet access from EC2 instance - Updates, downloads not working	- No Internet Gateway - Route table missing 0.0.0.0/0 route to IGW - Subnet not associated with correct route table	- Attach Internet Gateway - Add route 0.0.0.0/0 → IGW - Associate subnet with correct route table
 Cannot connect to instance (SSH/RDP)	- Connection timed out or rejected - Security group error	- Incorrect Security Group rules - NACL blocking the port - Wrong IP / DNS	- Check SG inbound rule - Check NACL inbound rule - Verify IP / DNS
 Instance cannot reach RDS	- Timeout connecting to database - App errors	- Security Group not allowing DB port - RDS security group doesn't allow EC2 - Route table issue	- Allow DB port in SG (e.g. 3306) - Allow source SG / CIDR in RDS SG - Verify route table
 Traffic not reaching correct destination	- Some services unreachable - Intermittent connectivity	- Wrong route in route table - More specific route missing - Overlapping CIDR	- Review route table - Add correct / specific route - Avoid overlapping CIDRs
 Access denied (even though route is correct)	- Connection reset or blocked - App unreachable	- NACL denying traffic - SG denies traffic - IAM / endpoint policy issues	- Check NACL rules - Check Security Group rules - Verify IAM / Endpoint policy

USEFUL TOOLS & COMMANDS

-  VPC Reachability Analyzer
→ Check connectivity between resources
-  Ping / Traceroute
→ Test network path
-  Security Group Flow Logs
→ See allowed / denied traffic
-  VPC Flow Logs
→ Monitor traffic at ENI / subnet level
-  AWS CLI
→ `aws ec2 describe-route-tables`
→ `aws ec2 describe-security-groups`

BEST PRACTICES TO AVOID ISSUES

- ✓ Use meaningful names and tags.
- ✓ Keep route tables simple and specific.
- ✓ Follow least privilege in SGs and NACLs.
- ✓ Use VPC Endpoints to avoid Internet dependency.
- ✓ Monitor with Flow Logs and CloudWatch.
- ✓ Document your network design.



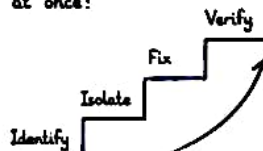
Prevention is always better than fixing!

TRAFFIC FLOW CHECKLIST

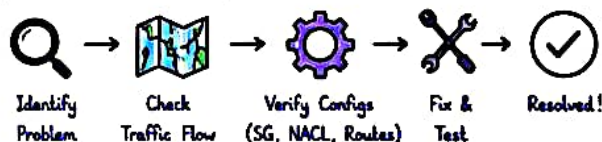
- ✓ Correct route in route table?
- ✓ Subnet associated with correct route table?
- ✓ Security Group allows the traffic?
- ✓ NACL allows the traffic?
- ✓ Network ACL rule order correct?
- ✓ Target resource is healthy?
- ✓ DNS resolution working?
- ✓ Where is the traffic getting blocked?

TIP: ★

Always test step-by-step. Don't try to fix everything at once!



QUICK DEBUG FLOW



A little troubleshooting today, a lot of uptime tomorrow!